

J-Delay

Four 1980s Japanese digital delays: chorus-delay, pitch-shift, compact and sampling.



Overview

J-Delay brings together four characters of the early Japanese digital delays of the 1980s - the 12- and 14-bit converters, modulated loops, pitch shifting in the feedback and sample loops. The four program keys change the algorithm.

VINTAGE DIGITAL is the flagship rack delay, with a dual-phase LFO (sine on the left, cosine on the right) for a wide chorus-delay. PITCH 12 is a 12-bit-style, 32 kHz unit with a pitch shifter in the loop, so every repeat moves by the PITCH interval.

COMPACT is the small unit: a darker loop (held under 8 kHz) and a compander that breathes, with its faint hiss. SAMPLE HOLD locks the last stretch of audio into a loop for stutters and phrase sampling.

In every program, changing the delay time glides the pitch as on the hardware, and the clock is never quite steady. The VINT key switches on the vintage converters - a lower sample clock and fewer bits - for the gritty, aliased sound of the originals.

Reach for it for

- Wide 80s chorus-delay on guitars, synths and vocals
- Cascading octave and fifth repeats (PITCH 12)
- Dark, breathing slapback and vocal echoes (COMPACT)
- Stutters and live phrase loops (SAMPLE HOLD)
- Lo-fi, bit-reduced repeats

Controls

Programs and status keys

VINTAGE DIG	The rack chorus-delay. The LFO runs sine on the left and cosine on the right, so MOD DEPTH spreads the repeats wide.
PITCH 12	Every repeat is pitch-shifted by PITCH semitones, so with feedback the echoes climb or fall. This program always uses the vintage converters, at 32 kHz and the BITS setting (12 by default).
COMPACT	The compact unit: the loop is held under 8 kHz (HI CUT can go no higher), and a 2:1 compander squeezes the signal into the loop and expands it coming out, so the repeats breathe, with a faint hiss.
SAMPLE HOLD	The sampling delay. With HOLD on, the last TIME L of audio plays round and round; nothing new is recorded.
SYNC	Locks TIME L and TIME R to the project tempo as note values: 1/1 to 1/32, with dotted (D) and triplet (T) values.
HOLD	Freezes the loop in SAMPLE HOLD. Switching it on from another program selects SAMPLE HOLD. The display shows HOLD in red while it is held.
VINT	The vintage clock: the delay line runs at the old hardware's sample rate with nearest-sample reads, and the converters use BITS bits. Off is the clean MODERN path. Always lit in PITCH 12.

Display, meters and presets

DELAY L / DELAY R	The delay times in milliseconds, in orange. The green line underneath shows FREE RUNNING, or the SYNC note values and tempo, and the converter setting (VINTAGE with its bits, or MODERN 24-BIT).
INPUT / DELAY	Level meters for the input and for the delayed signal.

BITS	Steps the vintage converters through 10, 12, 14 and 16 bits (12 by default). Fewer bits give more grit and noise. Used when VINT is on (and always in PITCH 12).
PRESETS	Opens the list of factory presets. The ◀ / ▶ keys beside it step through the ten presets, and the display beside them shows the number of the one in use. The last entry resets everything to the defaults.

The knobs

Each knob has a green readout underneath that shows its value while you touch it, and a red light that is lit whenever the knob is away from its default. Double-click a knob to reset it; hold Shift while dragging for fine moves.

TIME L / TIME R	The left and right delay times, 1 to 2700 ms (300 and 390 ms by default), or a note value with SYNC on. Changing the time glides the pitch, as on the hardware.
FEEDBACK	How much of the delay is fed back for more repeats, 0 to 99% (35% by default).
CROSS	Cross-feed, 0 to 100%: sends the feedback to the opposite side, so the repeats ping-pong.
LO CUT	Thins the repeats below this frequency, 20 to 1000 Hz (80 Hz by default). Each repeat gets thinner.
HI CUT	Darkens the repeats above this frequency, 1 to 20 kHz (9 kHz by default). Each repeat gets darker.
MOD DEPTH	How far the LFO sweeps the delay time, 0 to 10 ms (1.2 ms by default). Low settings give a gentle shimmer, high settings a deep chorus or vibrato.
MOD RATE	The LFO speed, 0.05 to 10 Hz (0.8 Hz by default).
PITCH	The interval each repeat is shifted by in PITCH 12, from -12 to +12 semitones (+12 by default). Labelled (PITCH) in the other programs, where it does nothing.

DUCK	Pulls the repeats down while you play, up to 18 dB, and lets them bloom back when you stop, 0 to 100%. Keeps busy parts clear.
RECOVERY	How quickly the repeats come back after ducking, 20 to 2000 ms (400 ms by default).
DRIVE	Drives the input stage before the delay, 0 to 100%, for a little grit.
WIDTH	The stereo width of the repeats, 0 to 150% (100% by default). 0 is mono; above 100% widens beyond the original.
MIX	Dry / delayed balance, 0 to 100% (30% by default).
LEVEL	Output level, -24 to +12 dB.

Drift

DRIFT	The clock is not quite steady: the left and right delay times wander slowly and independently, by up to a fraction of a millisecond, for a gentle, living pitch movement. 0 to 100% (50% by default): 50% is the standard amount, 100% twice that, 0 none; higher settings also wander a little faster.
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IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	Off by default. When on, it gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. An orange light shows while it is working.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

OFF / CLIP / LIMIT

The final stage: nothing, a soft clipper at -0.3 dBFS (no latency, the default), or a lookahead limiter holding the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). The OUT caption turns to a red CLIP while the clipper is working.

The header

S / M / L

Panel size: small, medium or large. The panel also scales to fit the plugin window.

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Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

80s Chorus Delay

VINTAGE DIGITAL at 300 / 390 ms with 1.8 ms of modulation: the classic wide chorus-delay.

Lead Guitar Echo

Synced 1/8 dotted repeats with 40% feedback, a little cross-feed and a darker top.

Dimension Wide

Very short 14 / 21 ms delays with deep, slow modulation and no feedback, for width.

Octave Cascade

PITCH 12 at 330 ms: each repeat climbs an octave.

Fifth Spiral

PITCH 12 with uneven times and cross-feed: each repeat rises a fifth and bounces side to side.

Octave Drop

PITCH 12 at 360 ms: each repeat falls an octave.

Dark Vocal Echo

COMPACT with synced quarter and dotted-quarter repeats, vintage clock on.

Compact Slap

COMPACT at 120 / 135 ms with light feedback: a short slapback.

Stutter Loop (press HOLD)	SAMPLE HOLD synced to a quarter note at 14 bits, 60% mix: press HOLD to loop it.
10-Bit Grit	VINTAGE DIGITAL on the vintage clock at 10 bits, with feedback and cross-feed.
Reset all to defaults	Every control back to its starting value.

Tips

- The presets set the program, times, feedback, filters, modulation, pitch, clock and mix; DRIVE, DUCK, WIDTH, DRIFT, LEVEL and the IN / OUT settings are left alone (except Reset).
- For live phrase sampling, choose SAMPLE HOLD, set TIME L to the length you want to catch (or use SYNC), then press HOLD at the end of the phrase.
- Automate TIME for tape-style pitch swoops: the time glides, so the repeats bend as it changes.
- Add DUCK on vocals and lead lines so the echoes stay out of the way while the part is playing and fill the gaps afterwards.

File: gfx-j-delay.jsfx